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# Blink completeness and rate in dry eye disease: An investigator-masked, prospective registry-based, cross-sectional, prognostic study

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ABSTRACT

*Purpose:* To investigate the prognostic ability of blink rate and the proportion of incomplete blinking to predict dry eye disease diagnosis, as defined by the TFOS DEWS II criteria.
*Methods:* A total of 453 community residents (282 females, 171 males; mean ± SD age, 37 ± 19 years) were recruited in an investigator-masked, prospective registry-based, cross-sectional, prognostic study. Dry eye symptomology, tear film quality, and ocular surface characteristics were assessed in a single clinical session, and blink parameters evaluated by an independent masked observer.
*Results:* Overall, 214 (47 %) participants fulfilled the TFOS DEWS II criteria for dry eye disease. Multivariate regression analysis demonstrated that an increased proportion of incomplete blinking was associated with a higher odds of dry eye disease (odds ratio, 1.12 per 10 % increase; 95 % CI, 1.05–1.19; $p < 0.001$), with the Youden-optimal prognostic threshold proportion being ≥ 40 % incomplete blinking. Higher levels of incomplete blinking were also associated with poorer dry eye symptomology, tear film stability, corneal and lid margin staining, lipid layer thickness, meibography, and meibum quality (all $p \leq 0.03$). No significant associations were detected between blink rate and ocular surface parameters (all $p > 0.10$).
*Conclusions:* The degree of incomplete blinking is a significant predictor of dry eye disease, and the utility of incorporating blink assessment into diagnostic workup algorithms warrants further investigation. The association with meibomian gland dropout, expressed meibum quality, and lipid layer thickness would suggest that incomplete blinking may predispose towards the development of meibomian gland dysfunction.

## 1. Introduction

The blinking mechanism plays a critical role in maintaining homeostasis of the ocular surface, tear film, and lacrimal functional unit [1–3]. The blink cycle fuels the lacrimal pump through the muscular action of the eyelids, and promotes delivery of the meibomian gland lipid secretions that inhibits aqueous tear evaporation and lubricates the ocular surface [1–6]. The blinking action also provides mechanical protection and debris clearance, as well as ensuring replenishment, mixing, and distribution of the tear film constituents over the ocular surface [1–3].

There has been growing recognition of the association between dry eye disease with pathological and physiological variations in blink dynamics [1,4,7–10]. An earlier study reported that the presence of incomplete blinking was associated with two-fold increased odds of developing dry eye disease in a cohort of community residents with no other systemic or ophthalmic comorbidities [9]. Previous studies have also highlighted that the diminished blink rate and increased tendency for incomplete blinking during tasks with high cognitive demand, such as digital screen use, can exacerbate dry eye symptoms and tear film instability [11–15]. A number of neurological and orbital conditions which adversely impact blink dynamics, including cranial nerve VII palsy, Parkinson's disease, progressive supranuclear palsy, and Grave's orbitopathy, have also been associated with meibomian gland dysfunction [16–19]. Conversely, preliminary prospective interventional studies have suggested that blink training may potentially improve clinical signs and symptoms of dry eye disease [2,12,20–22].

More recently, diagnostic instruments, such as the TearScience LipiView II Ocular Surface Interferometer (Johnson & Johnson Vision, California, United States), which provide automated measurement of

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blink rate and the proportion of incomplete blinking have become commercially available [7]. The purpose of this investigator-masked, prospective registry-based, cross-sectional, prognostic study was to evaluate the prognostic ability of blink rate and the proportion of incomplete blinking in predicting dry eye disease diagnosis, as defined by the global consensus Tear Film and Ocular Surface Society Dry Eye Workshop II (TFOS DEWS II) criteria.

## 2. Methods

### 2.1. Participants

This investigator-masked, prospective registry-based, cross-sectional, prognostic study adhered to the tenets of the Declaration of Helsinki, and was approved by the University of Auckland Human Participants Ethics Committee. The prospective registry is based at a single university centre, and recruits volunteers through open advertisement from local community residents based in the greater Auckland area. Participants were required to be at least 18 years of age; with no history of contact lens wear within the past two weeks; no history of major systemic (cardiovascular diseases, diabetes mellitus, dyslipidaemia, migraine headaches, thyroid disease, Crohn's disease, irritable bowel syndrome, lymphoma, malignancies), neurological, autoimmune (Sjögren syndrome, lupus erythematosus, rheumatoid and inflammatory arthritis, systemic sclerosis and scleroderma), orbital, or ophthalmic conditions (other than dry eye disease and meibomian gland dysfunction); no use of systemic or topical medications known to trigger dry eye disease; and no previous ophthalmic, oculoplastic or orbital surgery. Eligible participants were enrolled after providing written consent.

A total of 453 eligible participants were recruited, exceeding the sample size requirements for the desired study power. Correlation analysis power calculations (NCSS PASS 2002, Utah, USA) showed that a minimum of 347 participants were required to detect a standardised coefficient of 0.15, at 80 % power ($\beta = 0.2$) and a two-sided statistical significance level of 5 % ($\alpha = 0.05$).

### 2.2. Measurements

Participants were assessed at a single site, with a mean ± SD room temperature of 20.4 ± 0.6 °C and a mean ± SD relative humidity of 61.8 ± 6.6 %. Clinical measurements were performed on the right eye of each participant in a single session. Evaluation of blinking parameters was conducted by an independent, clinically-experienced observer who was masked to the results of all other ocular surface testing. Clinical measurements were conducted in accordance with the recommendations of the TFOS DEWS II Diagnostic Methodology subcommittee [7]. To minimise the impact on ocular surface and tear film physiology for subsequent assessments, clinical measurements were performed in ascending order of invasiveness [7], as listed in Table 1, and with ≥5 min gap between clinical tests. The diagnostic criteria for dry eye disease was based on the global consensus recommendations of the TFOS DEWS II Diagnostic Methodology report, as summarised in Table 2 [7].

**Table 1**
Order of clinical assessments conducted during the study visit.

| Assessments |
|---|
| OSDI dry eye questionnaire |
| DEQ-5 dry eye questionnaire |
| Blinking pattern assessment |
| Tear meniscus height |
| Non-invasive tear film breakup time |
| Tear film lipid layer interferometry |
| Tear osmolarity |
| Ocular surface staining |
| Expressed meibum quality |
| Infrared meibography |

The Ocular Surface Disease Index (OSDI) and 5-Item Dry Eye Questionnaire (DEQ-5) were administered to grade the level of dry eye symptomology [7].

Blinking pattern assessment and tear film lipid layer thickness evaluation were conducted with the LipiView II Ocular Surface Interferometer (TearScience Inc, North Carolina, USA). Automated objective readings of the blink rate and the proportion of incomplete blinking were obtained from the in-built proprietary blink dynamic module of the LipiView II programme, and averaged from three separate 20-second recordings. Tear film lipid layer thickness was evaluated using automated white light interferometry colour assessment by specular reflection, while the participant maintained fixation and blinked freely, as instructed. The average lipid layer thickness over a 20-second interval was recorded [23].

Tear meniscus height, non-invasive tear film breakup time, and tear film lipid layer grade were assessed using the Keratograph 5 M (Oculus Optikgeräte GmbH, Wetzlar, Germany). The lower tear meniscus height was assessed using pre-calibrated high-magnification digital imaging, and three measurements near the centre of the lower meniscus were averaged [7]. Non-invasive tear film breakup time was measured using automated detection of first break-up, while the subject maintained fixation and was requested to refrain from blinking. Three breakup time readings were averaged in each case [7]. Tear film lipid layer interferometry was graded according to the modified Guillon-Keeler system: grade 1, open meshwork; grade 2, closed meshwork; grade 3, wave or flow; grade 4, amorphous; grade 5, coloured fringes; grade 0, non-continuous layer (non-visible or abnormal coloured fringes) [24,25].

Tear film osmolarity measurements were performed with a clinical osmometer (TearLab, California, USA), from 50nL tear samples drawn from the lower lateral canthus tear meniscus. A measurement was taken for each eye, and the higher reading and the inter-ocular difference recorded [7].

Sodium fluorescein and lissamine green dyes were applied using the recommended technique described in the TFOS DEWS II Diagnostic Methodology report, in order to evaluate localised corneal, conjunctival, and lid margin areas of epithelial staining [7].

Meibum was expressed from the inferior eyelid using the Meibomian Gland Evaluator (TearScience, North Carolina, USA), with a standardised pressure of 1.2 g/mm$^2$, applied just inferior to the of the eyelid margin, and the quality of expressed meibum was graded according to the following scale: grade 0, clear fluid; grade 1, slightly turbid; grade 2, thick opaque; grade 3, toothpaste like; grade 4, complete orifice blockage [7,25].

Infrared meibography was imaged using the Oculus Keratograph 5M, with the superior and inferior eyelids everted in turn. From the captured image, the proportion of meibomian glands visible within the tarsal area was graded according to the five-point Meiboscale: grade 0, 0 %; grade 1, 1 % to 25 %; grade 2, 26 % to 50 %; grade 3, 51 % to 75 %; grade 4, 76 % to 100 % [7,26].

## 3. Statistics

Statistical analysis was performed with IBM SPSS version 29.0 (New

**Table 2**
Diagnostic criteria for dry eye disease based on the global consensus recommendations of the Tear Film and Ocular Surface Society Dry Eye Workshop II (TFOS DEWS II).[7].

| Diagnosis | Criteria |
|---|---|
| Dry eye disease | One of the following: OSDI score ≥ 13, or DEQ-5 score ≥ 6<br>AND<br>One of the following: non-invasive tear film breakup time < 10 s, tear osmolarity ≥ 308 mOsm/L, inter-ocular difference in osmolarity > 8 mOsm/L, corneal staining > 5 spots, conjunctival staining > 9 spots, or lid margin staining ≥ 2 mm length and ≥ 25 % width |

**Table 3**
Ocular surface parameters of participants. Data are presented as mean ± SD, median (IQR), or number of participants (% of participants). Asterisks denote statistically significant values ($p < 0.05$).

| Characteristic | All participants (n = 435) | Participants with dry eye disease (n = 214) | Participants without dry eye disease (n = 221) | p-value |
|---|---|---|---|---|
| **Dry eye disease diagnosis** | | | | |
| TFOS DEWS II criteria for dry eye disease diagnosis | 214 (47 %) | 214 (100 %) | 0 (0 %) | – |
| **Dry eye symptomology** | | | | |
| OSDI score (out of 100) | 23 ± 18 | 36 ± 23 | 10 ± 9 | <0.001* |
| DEQ-5 score (out of 22) | 9 ± 4 | 12 ± 4 | 5 ± 3 | <0.001* |
| **Tear film quality** | | | | |
| Non-invasive tear film breakup time (s) | 7.4 (5.1–10.8) | 6.1 (4.7–9.1) | 8.9 (5.6–12.3) | 0.005* |
| Tear film osmolarity (mOsmol/L) | 310 ± 16 | 311 ± 17 | 308 ± 15 | 0.03* |
| Inter-ocular difference in osmolarity (mOsmol/L) | 10 ± 8 | 12 ± 10 | 8 ± 7 | 0.01* |
| Tear film lipid layer grade (out of 5) | 3 (2–4) | 2 (2–3) | 3 (3–4) | 0.003* |
| Tear film lipid layer thickness (nm) | 71 ± 20 | 65 ± 19 | 76 ± 20 | <0.001* |
| Tear meniscus height (mm) | 0.24 ± 0.12 | 0.21 ± 0.11 | 0.26 ± 0.12 | <0.001* |
| **Ocular surface characteristics** | | | | |
| Corneal staining > 5 spots | 58 (13 %) | 40 (19 %) | 18 (8 %) | 0.002* |
| Conjunctival staining > 9 spots | 135 (30 %) | 94 (44 %) | 41 (19 %) | <0.001* |
| Lid wiper epitheliopathy ≥ 2 mm length and ≥ 25 % width | 259 (57 %) | 169 (79 %) | 90 (41 %) | <0.001* |
| Superior meibography grade (out of 4) | 1 (1–2) | 1 (1–2) | 1 (0–2) | 0.03* |
| Inferior meibography grade (out of 4) | 1 (1–2) | 1 (1–2) | 1 (0–1) | 0.01* |
| Expressed meibum grade (out of 4) | 1 (0–2) | 1 (1–2) | 1 (0–1) | 0.007* |

York, USA) and Graph Pad Prism version 8.01 (California, USA). Participants from the study cohort were randomised into developmental (80 %) and validation samples (20 %) for the purposes of constructing and evaluating a multivariable logit prognostic model for predicting dry eye disease diagnosis as defined by the TFOS DEWS II [7], using blink rate and proportion of incomplete blinking as predictor variables, and age, sex, ethnicity, and contact lens wear as pre-designated confounding variables. Age, sex, ethnicity, and contact lens wear are major risk factors for the development of dry eye disease which have been identified by the global consensus TFOS DEWS II Epidemiology Report to be corroborated by consistent evidence in the peer-reviewed scientific literature [27].

Multivariable logistic regression analysis was used to assess the relationship between blinking and ocular surface parameters, adjusting for age, sex, ethnicity, and contact lens wear as pre-designated confounding variables in the developmental sample. Blink rate, proportion of incomplete blinking, the interaction of blink rate and proportion of incomplete blinking, age, sex, ethnicity, and contact lens wear were entered as independent variables in the logistic regression analysis. Ocular surface parameters were dichotomised as dependent variables in the logistic regression analysis, using diagnostic thresholds defined by the TFOS DEWS II Diagnostic Methodology Report [7], or studies that evaluated optimal diagnostic cut-offs using the TFOS DEWS II criteria as the reference standard [28–30]. The Youden-optimal prognostic threshold proportion of incomplete blinking for predicting dry eye disease was then calculated.

The discriminative performance of the multivariable logit prognostic model for predicting dry eye disease diagnosis as defined by the TFOS DEWS II criteria was then evaluated in the internal validation sample with the area under the receiver operating curve (C-statistic). All tests were two tailed, and $p < 0.05$ was considered significant. Data are presented as mean ± SD, median (IQR), or number of participants (% of participants), unless otherwise stated.

## 4. Results

The mean ± SD age of the 453 participants (282 females, 171 males) was 37 ± 19 years (range, 18 to 92 years). Demographic and clinical

**Table 4**
Demographic and blink characteristics of participants. Data are presented as mean ± SD, or number of participants (% of participants). Asterisks denote statistically significant values ($p < 0.05$).

| Characteristic | All participants (n = 435) | Participants with dry eye disease (n = 214) | Participants without dry eye disease (n = 221) | p-value |
|---|---|---|---|---|
| **Demographics** | | | | |
| Age (years) | 37 ± 19 | 39 ± 20 | 35 ± 19 | 0.02* |
| Female sex | 282 (65 %) | 145 (68 %) | 137 (62 %) | 0.23 |
| Ethnicity | | | | |
| European ethnicity | 183 (40 %) | 60 (28 %) | 123 (56 %) | <0.001* |
| East Asian ethnicity | 225 (50 %) | 136 (64 %) | 89 (40 %) | <0.001* |
| South Asian ethnicity | 26 (6 %) | 15 (7 %) | 11 (5 %) | 0.42 |
| Other ethnicity | 19 (4 %) | 10 (5 %) | 9 (4 %) | 0.82 |
| Contact lens wear | 49 (11 %) | 30 (14 %) | 19 (9 %) | 0.09 |
| **Blinking parameters** | | | | |
| Blink rate ($minute^{-1}$) | 24 ± 13 | 25 ± 14 | 23 ± 13 | 0.42 |
| Proportion of incomplete blinking (%) | 58 ± 35 | 69 ± 34 | 46 ± 35 | <0.001* |

**Table 5**
Multivariable logistic, ordinal, or linear regression analysis of ocular surface parameters by blinking characteristics, adjusted for age, sex, and ethnicity in the developmental sample. Data are presented as odds ratios (95 % CI). Asterisks denote statistically significant values ($p < 0.05$).

| | Blink rate (per minute$^{-1}$) | | Proportion of incomplete blinking (per 10 %) | | Interaction of blink rate and incomplete blinking |
|---|---|---|---|---|---|
| Ocular surface parameter | Odds ratio (95 % CI) | p-value | Odds ratio (95 % CI) | p-value | p-value |
| TFOS DEWS II criteria for dry eye disease diagnosis | 1.00 (0.99–1.02) | 0.63 | 1.12 (1.05–1.19) | <0.001* | 0.87 |
| OSDI score ≥ 13 | 1.01 (0.99–1.03) | 0.24 | 1.09 (1.03–1.15) | 0.002* | 0.53 |
| DEQ-5 score ≥ 6 | 1.01 (0.99–1.03) | 0.48 | 1.05 (0.99–1.11) | 0.09 | 0.82 |
| Non-invasive tear film breakup time < 10 s | 1.00 (0.98–1.02) | 0.93 | 1.07 (1.02–1.13) | 0.007* | 0.36 |
| Tear osmolarity ≥ 308mOsmol/L | 1.00 (0.98–1.02) | 0.94 | 1.02 (0.94–1.06) | 0.67 | 0.29 |
| Inter-ocular difference in osmolarity > 8mOsmol/L | 1.00 (0.98–1.02) | 0.97 | 1.04 (0.96–1.09) | 0.34 | 0.18 |
| Corneal staining > 5 spots | 1.01 (0.99–1.03) | 0.37 | 1.23 (1.12–1.38) | <0.001* | 0.97 |
| Conjunctival staining > 9 spots | 1.00 (0.99–1.02) | 0.58 | 1.05 (0.99–1.10) | 0.16 | 0.84 |
| Lid wiper epitheliopathy ≥ 2 mm length and ≥ 25 % width | 1.01 (0.99–1.03) | 0.31 | 1.14 (1.07–1.20) | <0.001* | 0.66 |
| Tear meniscus height < 0.2 mm | 1.01 (0.99–1.03) | 0.22 | 1.05 (0.99–1.12) | 0.09 | 0.72 |
| Tear film lipid layer grade ≤ 3 | 1.00 (0.98–1.02) | 0.71 | 1.17 (1.09–1.32) | <0.001* | 0.51 |
| Tear film lipid layer thickness ≤ 72 nm | 1.00 (0.98–1.02) | 0.82 | 1.10 (1.04–1.17) | 0.001* | 0.37 |
| Superior meibography grade > 1 | 1.00 (0.98–1.02) | 0.87 | 1.08 (1.03–1.14) | 0.004* | 0.47 |
| Inferior meibography grade ≥ 1 | 0.99 (0.98–1.01) | 0.34 | 1.06 (1.01–1.12) | 0.03* | 0.75 |
| Expressed meibum grade ≥ 1 | 1.00 (0.99–1.02) | 0.53 | 1.15 (1.08–1.21) | <0.001* | 0.29 |

characteristics are presented in Table 3. Overall, 214 (47 %) participants fulfilled the TFOS DEWS II criteria for dry eye disease. The proportion of incomplete blinking was higher in participants with dry eye disease than those without (69 ± 34 % versus 46 ± 35 %, $p < 0.001$), although no significant inter-group differences were detected for blink rate (25 ± 14 versus 23 ± 13, $p = 0.42$). Table 4.

Multivariate regression analysis of ocular surface parameters by blinking characteristics in the developmental sample are summarised in Table 5. Logistic regression analysis, adjusted for age, sex, ethnicity, and contact lens wear, demonstrated that increased proportion of incomplete blinking was associated with higher odds of dry eye disease (odds ratio = 1.12 per 10 % increase; 95 % CI, 1.05–1.19; $p < 0.001$), with the Youden-optimal prognostic threshold proportion being ≥ 40 % incomplete blinking.

Higher levels of incomplete blinking were also associated with poorer OSDI scores, non-invasive tear film breakup time, corneal and lid margin staining, tear film lipid layer grade and thickness, superior and inferior meibography, and expressed meibum quality grades (all $p \leq 0.03$). No significant associations were detected between blink rate and ocular surface parameters, or the interaction of blink rate with incomplete blinking and ocular surface parameters (all $p > 0.10$).

In the internal validation sample, the C-statistic for the multivariable logit prediction model incorporating blink rate, incomplete blinking, and pre-designated confounding variables was 0.627 (95 % CI, 0.511–0.743; $p = 0.04$), and the accuracy was 0.648 (95 % CI, 0.546–0.739). The C-statistic for the model without blink rate and incomplete blinking was lower at 0.535 (95 % CI, 0.413–0.710; $p = 0.56$), and the accuracy was 0.560 (95 % CI, 0.458–0.658).

## 5. Discussion

To our knowledge, the current study is the first to investigate the predictive ability and optimal prognostic cut-off of the proportion of incomplete blinking to predict dry eye disease diagnosis, as defined by the global consensus TFOS DEWS II criteria. Multivariate regression analysis showed that there were 12 % increased odds of dry eye disease for every 10 % increase in the proportion of incomplete blinking, and the Youden-optimal prognostic threshold proportion was ≥40 % incomplete blinking. These findings are consistent with previous research which reports that the presence of incomplete blinking is associated with a higher risk of dry eye disease development in healthy community residents with no other systemic or ophthalmic comorbidities [9]. Moreover, earlier studies have also highlighted the link between dry eye disease with neurological and orbital conditions which adversely impact blink dynamics, such as cranial nerve VII palsy, Parkinson's disease, progressive supranuclear palsy, and Grave's orbitopathy [16–19].

In the current study, higher levels of incomplete blinking were also associated with poorer OSDI dry eye symptomology scores, non-invasive tear film breakup time, corneal and lid margin staining, lipid layer grading and thickness, superior and inferior meibography, and expressed meibum quality. These results would add further weight to the hypothesis that incomplete blinking increases the propensity of developing meibomian gland dysfunction and evaporative dry eye disease [1–5,7–10]. The blink cycle is thought to stimulate the lipid secretions of the meibomian glands through the contraction of the muscles of Riolan exerting pressure on the gland ducts [31,32]. The reduction in mechanical forces generated by the muscles of Riolan associated with incomplete blinking might potentially compromise the flow of lipid secretions in the ducts of the meibomian glands, thereby contributing to subsequent gland obstruction, dropout and atrophy [1–6,9]. The ensuing reduction in the integrity and quality of the tear film lipid layer secondary to meibomian gland dysfunction, as well as the increased inter-blink interval associated with incomplete blinking, can both trigger hyper-evaporation of the tear film and desiccation of the exposed ocular surface [2,3,5,9,33]. This can be further exacerbated by the inefficient distribution and mixing of meibomian lipids in the tear film associated with incomplete blinking, which also acts to compromise the integrity of the surface lipid layer [2,3,5,34]. The resulting tear film instability can then trigger a vicious circle of ocular surface inflammation, that can exacerbate pre-existing clinical signs and symptoms of dry eye disease [3,35].

In the current study, no significant associations were detected between blink rate and ocular surface parameters. Inconsistent findings have been reported in previous research, with some studies suggesting that decreased blink rate is associated with poorer tear film stability [36,37], while another study also reported no clear correlation [9]. However, it is noted that these trends might be partially confounded by the involuntary compensatory reflex mechanisms which increase blink rate secondary to ocular surface dryness [9,38], and might contribute to the lack of apparent association between blink rate and ocular surface parameters.

The results of this study showed that the proportion of incomplete blinking was a significant predictor of dry eye disease, and the utility of incorporating of blink assessment into diagnostic workup algorithms

would warrant further investigation [7,39]. Indeed, the identification of patients exhibiting higher degrees of incomplete blinking might help to tailor multimodal approaches to the prevention and treatment of dry eye disease, including the incorporation of blink training [2,12,20–22,40,41]. However, it is noted that in the internal validation sample of the current study, the accuracy of the multivariable logit prediction model incorporating blink rate, incomplete blinking, and pre-designated confounding variables was 0.648. The modest accuracy of the multivariable model might partially reflect the highly heterogenous and multifactorial nature of dry eye disease, as well as the discordance between clinical signs and symptoms [7].

This study is not without limitations. The eligibility criteria required participants to have no previous ophthalmic, oculoplastic or orbital surgery, which might limit the applicability of study findings to patients with iatrogenic conditions secondary to these causal factors [3,42,43]. Moreover, patients with major systemic, neurological, autoimmune, orbital conditions were also excluded [16–19], although this was intentional to facilitate isolated investigation of the relationship between physiological variation in blink dynamics and ocular surface parameters in the current study. The assessment of blink rate in an artificial situation within the LipiView instrumentation rather than a natural seeing environment is acknowledged to be a study limitation, and it is noted that stronger association with other ocular surface parameters were reported in a previous study where blink rate was evaluated under natural seeing conditions [44]. The cross-sectional study design is acknowledged to preclude the inference of causal relationships, and the potential for subtle changes in the eyelid margin secondary to meibomian gland dysfunction to impact on blink completeness cannot be entirely discounted.

In conclusion, the results of this study showed that the proportion of incomplete blinking is a significant predictor of dry eye disease. Future studies are required to further investigate the potential role of incorporating blink assessment into dry eye prognostic and/or diagnostic workup algorithms, and whether patients exhibiting higher degrees of incomplete blinking might potentially benefit from multimodal approaches to the prevention and treatment of dry eye disease that incorporate blinking training. The association between incomplete blinking with meibomian gland dropout, meibum quality, and lipid layer thickness lends further support to the hypothesis that incomplete blinking may predispose towards the development of meibomian gland dysfunction.

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## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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